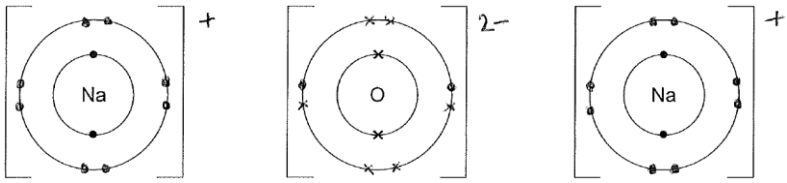
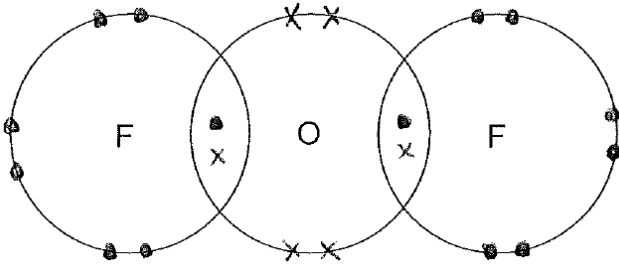


HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		 award (1) for 8 dots in outer shell of both sodium ions award (1) for 6 crosses and 2 dots in outer shell of oxide ion do not accept if extra electrons drawn in inner shell award (1) for + charge on sodium ions and 2– charge on oxide ion accept if + charge shown on only one sodium ion		3		3		
		(ii)		$4\text{Na} + \text{O}_2 \longrightarrow \boxed{2} \text{Na}_2\text{O}$ award (1) for product award (1) for balancing only if product correct		2		2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		 award (1) for two shared pairs award (1) for three octets accept shared electrons drawn on lines or in overlap award (1) if diagram wholly correct except using all dots or crosses		2		2		
		(ii)		bonds within the molecules are strong☐ bonds within the molecules are weak☐ bonds between the molecules are strong☐ bonds between the molecules are weak☒	1			1		
				Question 4 total	1	7	0	8	1	0